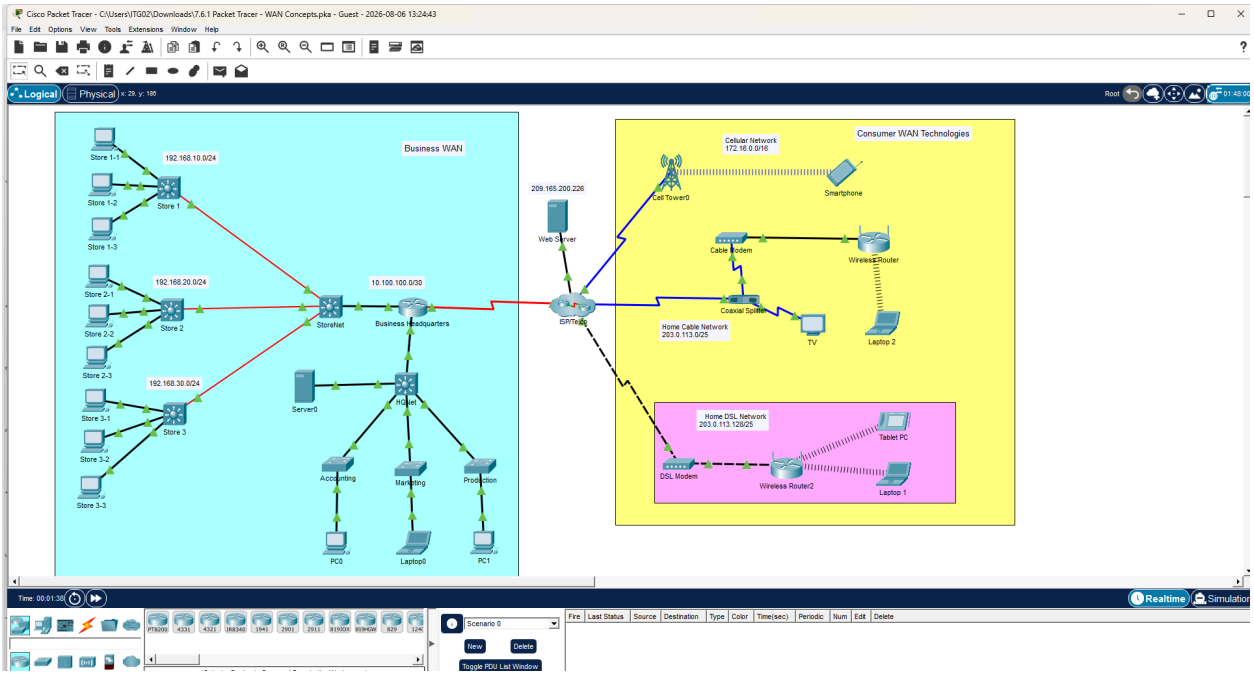
Lab Report: Packet Tracer - WAN Concepts



1. Objectives

- Investigate various types of WANs (Wide Area Networks) including consumer technologies (Cable, DSL, Cellular) and business WAN architectures.
- Analyze connectivity options and media types.
- Verify end-to-end connectivity and security separation.

2. Network Topology and Addressing

Note: This activity is an observation-based investigation. No custom router configurations or static address assignments were required, as devices utilize DHCP for addressing.

Topology Summary

- **Consumer WAN:** Comprises two home networks (Cable and DSL) and a cellular network.
- **Business WAN:** Comprises a retail tire store infrastructure with a local StoreNet switch connecting multiple store segments to a Business Headquarters router.
- **Connectivity:** The Business Headquarters router connects to the ISP via a serial WAN connection.

3. Investigation Findings (Consumer WAN)

- **Technologies:** The home networks use Cable and DSL WAN technologies.
- **Media:** The Cable network uses coaxial media connected via a splitter. The DSL network uses phone cable connected to a DSL modem.
- **DSL Modem:** Converts telephone data network signals to Ethernet. Ports: One connects to the phone line (Telco), one to the home LAN (Ethernet).
- **Splitter:** Necessary in the Cable network to separate digital data signals (for the cable modem) from video signals (for the TV).
- **Cable Modem:** Converts cable data signals to Ethernet. Connects to coaxial cable from the splitter and UTP cable from the Ethernet interface.
- **Smartphone:** IP address is 198.51.100.100 (assigned via DHCP from the Telco network). It is currently using cellular data.

4. Business WAN Analysis

- **Connections:** The network utilizes Ethernet (copper and fiber) and serial connections.
- **StoreNet Switch:** Contains Gigabit Ethernet copper ports and four modular ports with GLC-LH-SMD fiber optic SFP modules to facilitate connections over longer distances.
- **WAN Service:** The Business Headquarters router uses a serial WAN connection to connect to the ISP.

5. Connectivity Testing

- **Observation:** Hosts in both networks can ping the web server. However, hosts on the Business WAN networks cannot ping hosts on the Consumer WAN networks, and vice versa.
- **Conclusion:** This is a desirable security configuration, as these networks should not be directly reachable from external, unauthorized networks.